Lever Test · Tier Pack · Independent

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Independent Route (Book First)

Record the results for 5 cm (8 coins), 10 cm (4 coins), and 20 cm (2 coins), then write a conclusion quoting the numbers and naming one control variable.

Help: At 5 cm it took __ coins, while at 20 cm it took __ coins. This shows that as effort distance increases, the effort needed __.

Optional Make It · Independent

Make a 3-bar Effort Graph Strip

Draw three simple bars (8, 4, and 2 squares high) for 5 cm, 10 cm, and 20 cm on a graph slip and stick or copy it beside your conclusion in your book.

Word Help

effort distance: the distance from the pivot to where the effort push is applied

load: the weight or mass being lifted by the lever

independent variable: the one thing we change on purpose (the effort distance from the pivot)

pattern: how one measurement changes when we change another

Exit Check

Which effort distance needed the fewest coins: 5 cm or

20 cm?

What happens to the effort needed as you push further from the pivot?